



**המכללה האקדמית תל אביב יפו  
בית הספר למערכות מידע**

**Human-Computer Interface, 273020**

**ד"ר משה ליבה**

**תשפ"ה, סמסטר א', מועד א', 26/02/2025**

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**מספר זהות:**

**משך הבחינה: שעותיים**  
**חומר עזר: 3 דפי עזר (6 עמודים) ע"י הסטודנט, מילון**

בבחינה 27 שאלות, יש לענות על כולן. משקל השאלות זהה - 3.7 נקודות לשאלה.  
בכל שאלה יש לסמן את התשובה הנכונה ביותר (אחת בלבד).  
קרא בעיון את שלושת התרחישים לפני שתענה על שאלות מבוססות תרחישים.

**בהצלחה!**





## **Part A: General HCI (Questions 1-16)**

### **Question No. 1 (3.7 pts.)**

**When designing a voice-based virtual assistant, the biggest UX challenge is often:**

- a. Handling natural language inputs and providing clear spoken feedback.
- b. Maintaining consistent color palettes.
- c. Avoiding brand references in marketing copy.
- d. Using traditional mouse-click interactions.
- e. Keeping the code size minimal.

### **Question No. 2 (3.7 pts.)**

**In a heuristic evaluation, experts typically:**

- a. Check the interface against known usability best practices (e.g., Nielsen's).
- b. Conduct large-scale user surveys.
- c. Fix code-level errors in real time.
- d. Focus only on visual design consistency.
- e. Ignore minor usability issues entirely.

### **Question No. 3 (3.7 pts.)**

**A key advantage of low-fidelity prototypes is that they:**

- a. Allow for quick, cost-effective validation of initial design ideas.
- b. Eliminate the need for further user testing.
- c. Accurately simulate the final product's look and feel.
- d. Require advanced programming knowledge.
- e. Are only helpful at the end of development.

### **Question No. 4 (3.7 pts.)**

**In microcopy design (e.g., button labels, system messages), which principle is most crucial?**

- a. Use short, action-oriented phrases that clarify the next step.
- b. Keep text highly technical to demonstrate expertise.
- c. Provide minimal explanation to reduce visual clutter.
- d. Focus on formal language rather than user-friendly terms.
- e. Never reference user goals or context.

### **Question No. 5 (3.7 pts.)**

**Which choice best summarizes why designers should integrate accessibility from the earliest stages?**

- a. It allows more inclusive solutions that accommodate diverse user needs.
- b. It guarantees users will prefer darker interface themes.
- c. It reduces marketing campaign costs.
- d. It keeps the development timeline shorter by cutting user feedback.
- e. It is only relevant in government-regulated environments.





**Question No. 6 (3.7 pts.)**

**Which accessibility feature is particularly helpful for color-blind users?**

- a. High-contrast color combinations or icon-based indicators.
- b. Randomized color schemes on page load.
- c. Detailed audio descriptions for images.
- d. Minimizing all text in the interface.
- e. Complex animations that emphasize elements.

**Question No. 7 (3.7 pts.)**

**What is often the best method for evaluating an interface's accessibility for users with disabilities?**

- a. Conducting user testing with people who have diverse accessibility needs.
- b. Solely relying on the accessibility needs checklist.
- c. Automated load-time performance testing.
- d. Relying on color palettes matching strict brand guidelines.
- e. Letting only developers decide through code reviews.

**Question No. 8 (3.7 pts.)**

**Which of the following best describes a key difference between heuristic evaluation and usability testing?**

- a. Heuristic evaluations are performed by experts applying usability principles, while usability tests involve real users.
- b. Heuristic evaluations require large user samples, while usability tests do not.
- c. Both methods always produce the same findings.
- d. They are the same in terms of methodology.
- e. Heuristic evaluations measure brand awareness, while usability tests measure code quality.

**Question No. 9 (3.7 pts.)**

**In which scenario would a heuristic evaluation be most advantageous?**

- a. When you need quick expert feedback based on known usability principles, early in development.
- b. When you have a large, diverse group of end users ready to test.
- c. When measuring click-through rates on ad variants.
- d. When optimizing back-end code for minimal memory usage.
- e. When evaluating user opinions about brand marketing.

**Question No. 10 (3.7 pts.)**

**Which scenario is better suited for usability testing rather than a heuristic evaluation?**

- a. Verifying user behavior with a functioning prototype.
- b. Evaluating initial design sketches using expert judgment only.
- c. Checking compliance with standard heuristics.
- d. Ensuring consistent labeling across the interface.
- e. Identifying coding errors in real time.





**Question No. 11 (3.7 pts.)**

**How does combining heuristic evaluation and usability testing benefit a UX project?**

- a. It merges expert critique with real user feedback for a more complete view.
- b. It doubles the workload with no meaningful improvement.
- c. It eliminates the need to recruit representative users.
- d. It entirely replaces iterative design processes.
- e. It focuses primarily on system architecture and performance.

**Question No. 12 (3.7 pts.)**

**During requirements gathering, which is the main advantage of combining field observation (ethnography) with prototyping?**

- a. Ethnographic findings reveal real-world tasks and contexts, which prototypes then validate with users.
- b. Prototyping replaces the need for any field observation.
- c. Ethnography alone is sufficient for design and testing.
- d. Prototyping should happen before any user observation.
- e. Ethnography cannot inform prototyping because they are separate processes.

**Question No. 13 (3.7 pts.)**

**Which design practice would *not* support users with cognitive or learning difficulties?**

- a. Requiring users to memorize complex command sequences before actions.
- b. Simplifying on-screen tasks to reduce memory load.
- c. Providing multiple, clear ways to encode information (icons, color labels, text).
- d. Maintaining consistent layouts and recognizable symbols.
- e. Incorporating straightforward, guided steps throughout the interface.

**Question No. 14 (3.7 pts.)**

**An ATM interface emphasizes certain usability factors more than others. Which set is typically *most* crucial?**

- a. Errors, Efficiency, and Learnability.
- b. Strictly Satisfaction and Memorability.
- c. Only Errors and Memorability.
- d. Overall blend of all five usability aspects.
- e. Only Efficiency.

**Question No. 15 (3.7 pts.)**

**Now consider a 3D racing video game interface. Which aspects of the five usability factors are typically *most* important?**

- a. A balanced mix of Learnability, Memorability, Errors, Satisfaction, and Efficiency.
- b. Exclusively Errors and Learnability—gamers rarely care about fun.
- c. Only Efficiency—fast reflexes matter most.
- d. Strictly Memorability—players must memorize all combos.
- e. Satisfaction is irrelevant for entertainment.







**Question No. 16 (3.7 pts.)**

**When analyzing an e-commerce homepage, which clue indicates you are dealing with a *taxonomy* rather than an *ontology*?**

- a. You see labeled categories (e.g., "Electronics,") without explaining deeper relationships.
- b. You are defining the philosophical meaning of each product label.
- c. You are creating advanced user flows for every possible path.
- d. You are comparing performance metrics across prototypes.
- e. You are ignoring any attempt to organize content.

**Part B: Scenario 1: Conversational Banking App**

A voice-enabled banking app lets customers check balances, make payments, and request transaction histories via spoken commands. During testing, these issues emerge:

- Users hesitate to trust the voice assistant with sensitive transactions.
- The assistant struggles with different accents or background noise.
- Ambiguous responses cause confusion (e.g., "All set!" without clarifying the amount or recipient).

**Answer Questions 17-20 based on this scenario.**

**Question No. 17 (3.7 pts.)**

**Which usability principle most directly addresses the need for clear, unambiguous responses from the assistant?**

- a. Visibility of system status
- b. Aesthetic and minimalist design
- c. User control and freedom
- d. Error prevention
- e. Match between system and real world

**Question No. 18 (3.7 pts.)**

**Which action is likely to best address users' trust concerns over sensitive transactions?**

- a. Provide thorough confirmation prompts and restate transaction details
- b. Hide all security notices to reduce perceived complexity
- c. Use ambiguous confirmations like "All set!"
- d. Rely on a single-step, silent authentication method
- e. Eliminate all authentication steps to streamline the process

**Question No. 19 (3.7 pts.)**

**What is a good design approach for users with hearing or speech impairments in a voice-based interface?**

- a. Show on-screen text transcripts for all voice interactions
- b. Reduce the number of user options
- c. Force reliance on third-party captioning tools
- d. Remove any visual prompts to emphasize voice commands
- e. Hide alternative text-based interactions in advanced settings





**Question No. 20 (3.7 pts.)**

**Which usability test metric would be most appropriate for measuring success/failure in a voice-based app?**

- a. Task completion rate for key voice commands
- b. Number of times the app is downloaded
- c. Voice assistant's pitch and tone rating
- d. Lines of code executed per interaction
- e. Marketing budget spent on voice features

**Part C: Scenario 2: Retail Website Navigation Overhaul**

A large retail site is cluttered with thousands of product categories. Users rely heavily on the search bar, and session abandonment is rising. The UX team plans to reorganize the navigation and category structure.

**Answer Questions 21-24** based on this scenario.

**Question No. 21 (3.7 pts.)**

**Which method is most effective for discovering how users naturally group or label products?**

- a. Card sorting
- b. Eye-tracking studies
- c. Measuring ad click-through rates
- d. Building a high-fidelity prototype first
- e. Heuristic evaluation by a single expert

**Question No. 22 (3.7 pts.)**

**Consistent labeling (e.g., using "Shirts" instead of "Upper Wear" or "Tops") primarily supports which principle?**

- a. Consistency and standards
- b. Visibility of system status
- c. Error prevention
- d. Flexibility and efficiency of use
- e. Recognition rather than recall

**Question No. 23 (3.7 pts.)**

**During a cognitive walkthrough of the new navigation, which step would be least relevant to assess?**

- a. The color code used in the website's source code
- b. How a first-time user discovers the product categories
- c. Whether users can easily return to the homepage
- d. How a user moves from a product page to the checkout process
- e. Whether the user sees clear progress indicators





**Question No. 24 (3.7 pts.)**

**When running an A/B test on the new site navigation, which KPI is most relevant to measure success?**

- a. Percentage of abandoned sessions or bounce rate
- b. Number of staff hours used for each variant
- c. Lines of CSS changed in each version
- d. Developer satisfaction rating
- e. Total number of SKUs in the inventory

**Part D: Scenario 3: Mobile Fitness App Performance & Error Handling**

A mobile fitness app logs workouts, provides health tips, and syncs with wearables. Users complain about random crashes and cryptic error messages like "Unknown exception code: 4004" when syncing fails.

**Answer Questions 25-27 based on this scenario.**

**Question No. 25 (3.7 pts.)**

**Which negative consequence is most likely when error messages are cryptic?**

- a. Users feel confused and lose confidence in the app
- b. Users find the app less entertaining
- c. Users gain shallow insight into technical architecture
- d. Users feel motivated to fix the bug themselves
- e. No negative consequence occurs

**Question No. 26 (3.7 pts.)**

**Which improvement aligns best with microcopy best practices for these error messages?**

- a. Provide a clear, specific description and steps to fix the issue
- b. Keep messages vague, e.g., "Try again later."
- c. Show random memes to lighten the tone
- d. Display only numeric error codes for reference
- e. Remove error messages and handle them silently

**Question No. 27 (3.7 pts.)**

**Which of the following is an example of a measure to understand user satisfaction when crashes occur?**

- a. A post-task questionnaire asking about frustration or confidence
- b. The average time per workout session
- c. The ratio of successful syncs to failures
- d. The total CPU usage while the app runs
- e. The number of times the app is opened daily

